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Impact of a digital platform and flare risk blood biomarker index on lupus: A study protocol design for evaluating self efficacy and disease management

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ABSTRACT

We report the design of a pilot study employing clinic visits and a synergized digital, health coaching, and biomarker platform (aiSLE® MGMT [Management]) aimed at improving patient self-efficacy and patient-physician interactions to mitigate heightened disease activity, curtail clinical flares, and reduce potential long-term organ damage in systemic lupus erythematosus (SLE). This 12-month study will utilize a longitudinal cross-over design. Adult participants with confirmed SLE will be recruited and appropriately consented at five community-based US rheumatology clinics. Board certified rheumatologists will perform an exam, documenting the SLE Physician Global assessment (PGA) at Baseline, 3-, 6- and 9-month visits. Blood samples will be collected for completion of a newly developed and validated Lupus Flare Risk Index (L-FRI) that predicts the risk of developing a clinical flare in the next 12 weeks. The digital component of the study will be initiated following the 3-month visit. Participants will be given a mobile study app interfaced with a smartwatch to record activity, heart rate, and sleep quality. The app will be used to administer patient-reported outcome (PRO) surveys and health coaching. In the first three months participants and their rheumatologists will be blinded to L-FRI results; all L-FRI and digital results will be available starting in month four. This prospective pilot study will assess the impact of a comprehensive disease management platform employing a novel blood biomarker test, a mobile app interfaced with a smartwatch to collect biometric data, and app-based video health coaching on self-efficacy, clinical decisions, and outcomes in SLE disease management.

1. Introduction

Systemic autoimmune diseases, including SLE, afflict a significant proportion of the US population. Population-based studies in the US reflect a SLE prevalence of 73 per 100,000 persons with ranges from 42 to 300 cases per 100,000 persons across different studies [1–4] with an estimated annual economic burden on the US healthcare system of $31 billion [4]. SLE classification is contingent on meeting 4 of 11 American College of Rheumatology (ACR) [5,6] or 4 of 17 Systemic Lupus International Collaborating Clinics (SLICC) [7] criteria. Prevalence is higher in women of childbearing age (>90 %) and minority populations [8]. Persistently active clinical disease, disease flares, and their treatments place patients at risk for permanent organ damage [9–11], including nervous system, pulmonary, cardiovascular, and renal damage. Patients with waxing/waning disease and clinically active or quiescent disease are each at risk of clinical disease flare [12,13].

Optimal disease management of SLE to minimize disease activity and curtail flares that result in permanent organ damage is difficult, in part due to a persistent shortage of rheumatologists, both in the US and abroad [14]. It is therefore necessary to prioritize in-person clinic visits for those patients at greatest risk of heightened disease activity and flare, while still optimally managing lower risk SLE patients. Early intervention to prevent tissue and organ damage is challenging, as signs and symptoms of high disease activity and clinical disease flare are captured after their occurrence. Despite validated clinical disease activity instruments [15–17] and improved treatment strategies, SLE patients still

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suffer end-organ damage resulting in significant morbidity and early mortality [18] and an average of 1.8 disease flares per year [19]. Requiring more medical care and hospitalization, this results in average annual healthcare costs in excess of $20,000 (inpatient, outpatient, and pharmacy) [20]. Increased morbidity and early mortality associated with the frequency/severity of increased disease activity and flare [21] further escalate costs. Long-term use of steroids [22] and other immune-suppressants [23] required to manage disease activity are associated with increased morbidity. Novel tools are needed for proactive strategies to add or increase steroid-sparing immune modifying agents [24] to prevent end-organ damage [9–11] and reduce the pathogenic and socioeconomic burdens of SLE [4,25].

Clinical visits alone (every 3–12 months, depending on rheumatologist availability and patient geographical location), including medical evaluation and standard laboratory testing, do not accurately capture the impact of physiological, environmental, and psychological factors related to SLE and other chronic diseases. Patient participation and self-efficacy in disease management have been found to improve outcomes [26–29]; mounting evidence supports the utility of wearables interfaced with mobile phone apps for monitoring symptoms and promoting positive self-management behaviors in management of chronic conditions such as diabetes [30,31], cardiovascular disease [32,33], Parkinson's disease [34] and rheumatoid arthritis [35]. The impact of digital technologies has been enabled by the increased use and flexibility of mobile phones for delivering telehealth medical intervention. Health coaching provides professional support and expertise to assist patients to modify and implement lifestyle changes based on the information provided by digital tools. Overall, by enabling essentially continuous monitoring and feedback to patients, these digital tools have the potential to significantly advance disease management and improve patient outcomes.

This pilot study will assess the impact of a comprehensive platform employing a novel blood biomarker index, a mobile app interfaced with a smartwatch to collect biometric data, and mobile app-based health coaching. Our hypothesis is that this comprehensive platform will improve metrics of patient self-efficacy, clinical disease management, and outcomes in SLE by filling the existing gap between rheumatology clinic visits in the framework of SLE disease management (Fig. 1). The primary objective of the study is to evaluate the effect of the comprehensive platform on patient health self-efficacy, defined as an individual's confidence in managing their health condition and associated problems. The secondary objective is to evaluate the usefulness and impact of the comprehensive platform to rheumatologists in managing patients with SLE. The exploratory objective is to determine if the rheumatologist's perceptions of the patient's clinical status measured by the PGA and rheumatologists' surveys correlate with patients' immunological status as determined by their biomarker index scores.

This paper presents the design and methodology for this pilot study employing both in-person clinic visits and a comprehensive biomarker and digital platform for SLE disease management.

2. Methods and analysis

2.1. Study design

Fig. 1 presents an overview of the study showing the relationship between the major components, including the in-person interactions and flow of information. The participants will have in-person clinical visits at baseline and throughout the study. The digital components of the study involve wearable and health coach monitoring to capture physiological, environmental, and mental factors related to SLE on a regular basis and report the results to the participant's rheumatologist. Beginning in month four, monthly reports with a cumulative summary of the biomarker index results along with monthly summaries of patient-reported symptoms, quality of life (QOL) PRO survey results, and biometric data will be sent to the rheumatologist between clinic visits.

Fig. 2 shows an overview of the details of the study design. This is a 12-month, self-controlled longitudinal study (in which each participant serves as his or her own control over the first three months of the study). The study includes participant clinic visits and evaluations at baseline, 3-, 6-, and 9-months. Digital data collection (wearable), use of the app (including PRO surveys), and access to a certified health coach (through the app) will begin in month four and continue through the end of the study in month 12.

Rheumatologists will complete a start-of-study survey for each participant and an end-of-study survey (the content of surveys and evaluations are discussed in Section 2.4 below). Rheumatologists will

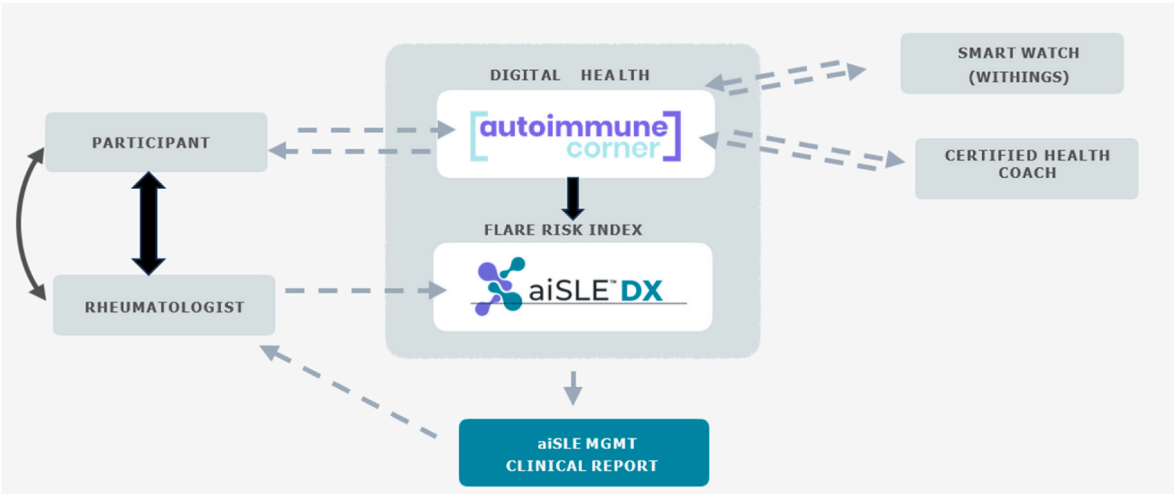

Fig. 1. Overview of Interactions and Flow of Information in the Study. The study participants and rheumatologists will have in-person clinic visits at 3-month intervals. Beginning in month 4, the participants will have access to the [AutoimmuneCorner]™app, smartwatch, and a certified health coach. The solid line with arrows shows interactions between rheumatologists and participants during clinic visits. The solid curved line with arrows shows in person care interactions between the rheumatologists and participants. The dashed lines show the flow of information in the study. The participant has access to smart watch collected information and the health coach through the app. The provider receives monthly summary reports summarizing biometric wearable data, PRO survey data and the L-FRI clinical reports. The provider reports these results to the participant (solid straight line with arrows). The dashed lines with arrows show the flow of information in the study. The curved solid line with double arrows shows the gap in information flow between visits that is filled by the aiSLE MGMT clinical report provided to the rheumatologists.

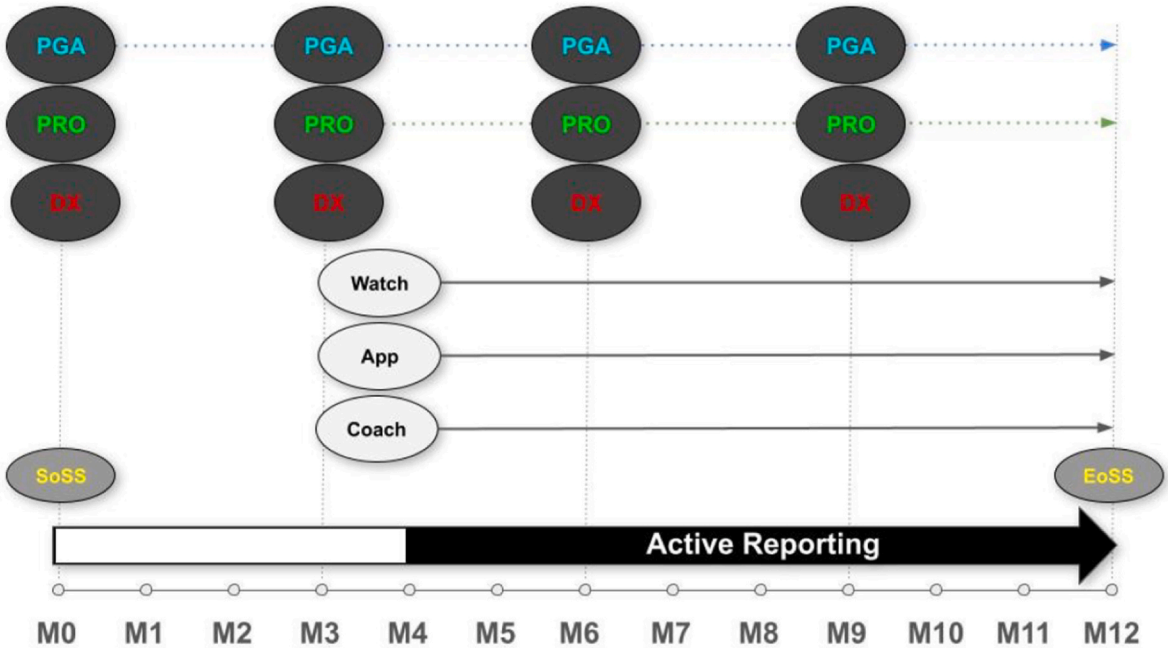

Fig. 2. Overview of the Flare Risk Biomarker and Digital Platform Study Over the Twelve-Month Study Period. In the diagram, "PGA" refers to quarterly participant exam, plus a rheumatologist's global assessment of SLE disease severity and activity, "PRO" reflects quarterly participant-reported outcome questionnaires, "DX" refers to biological sampling for biomarker quantitation, "SoSS" reflects start-of-study surveys completed by participants and rheumatologists, "EoSS" reflects end-of-study surveys completed by participants and rheumatologists, and "Watch", "App" and "Coach" refer to participant receipt of smart watches (for biometric tracking), digital apps (to facilitate PRO reporting), and access to a certified health coach. The reporting arrow indicates that, after the initial pre-intervention quarter, active reporting (facilitated by Watch, App and Coach) begins and continues for the duration of the study. The dotted line spanning the PGAs indicates that intermittent medical assessments may be available in appropriately consented medical records available during the study, and up to 24 months prior. The dotted line spanning the PROs indicates that, in addition to general quarterly surveys, the App enables real-time participant self-reports of medically notable circumstances including possible flares and SLE-associated symptoms.

complete an SLE-specific PGA at each clinic visit as a reflection of overall disease severity and activity. Participants will complete a start-of-study survey and end-of-study survey in month 12.

During the first three months of the study, participants will receive current standard of care treatment while biomarker index data are being collected but not reported to the rheumatologist or the participant. Starting in month 4 and continuing through month 12, participants will begin to use the study mobile app interfaced with a smartwatch, and SLE-specific health coaching with a professionally trained and certified coach will be available on the mobile app. The participants will be instructed to continuously wear the smartwatch both day and night and will have real time continuous access to their physiological metrics including activity (daily steps, exercise), awake and sleeping heart rate, and sleep quality (duration, depth, regularity/interruptions, breathing disturbances, snoring) from the Withings smartwatch. They will have access to informational modules on health and lifestyle factors in the app (described in detail in Section 2.4 below) as well as the opportunity to discuss and ask questions during the sessions with the certified health coach and in between sessions by messaging the health coach in the app.

Beginning in month 4 rheumatologists will receive blood biomarker index results from baseline and 3-month visits and will receive future results from the 6- and 9-month clinic visits upon completion. In addition, through the remainder of the study beginning in month 5 the rheumatologist will receive a monthly comprehensive report, with a cumulative summary of the biomarker index results along with monthly summaries of patient-reported symptoms, quality of life (QOL) PRO survey results, and biometric data. Summary reports will be transmitted by fax to each rheumatologist. The rheumatologists will be encouraged to share the data in these reports with the participants during in-person clinic visits and between visits as they judge necessary based on the information in the report.

The Western Institutional Review Board (WIRB) - Copernicus Group Institutional Review Board (WCG IRB) has approved this study protocol (Protocol #20222161), consent forms and all participant recruiting materials and survey materials.

2.2. Recruitment

The study design calls for a total of five board certified rheumatologists from five different geographically distinct clinics in the US to recruit participants to the study, with each site to identify and enroll 10 patients confirmed to have classified SLE for a total enrollment of 50. Candidate sites will be engaged by presenting a brief outline of the study and requesting a 15 to 30-min introductory meeting to provide an overview of the study, answer any questions about the study, and determine their interest moving forward with study participation. A second meeting will be set up with the interested sites to present a more detailed discussion of the study protocol, the requirements of the site, and logistics for sample collection/shipping. Five clinical sites have been identified at which the study is currently in progress. Participating clinics will be provided with IRB approved recruitment handouts and flyers. Rheumatologists and/or their staff will discuss the study and offer enrollment to their patients that meet the inclusion criteria during the regularly scheduled office visits.

2.3. Inclusion and exclusion criteria

Participants must have a clinical diagnosis of classified SLE as defined by meeting at least four ACR classification criteria for SLE [5,6] OR meeting at least four SLICC classification criteria for SLE [7], be at least 18 years old, and have had no biologic therapy in the last 12 months, defined as treatment with belimumab, rituximab, or similar therapies [36]. Belimumab is the only biological therapy FDA approved for the treatment of SLE and is a human monoclonal antibody that

inhibits the binding of soluble B lymphocyte stimulator (BLyS) to B cells. Treatment with anti-CD19 or anti-CD20 B cell depletion therapy (e.g. rituximab, ocrelizumab, obinutuzumab, or similar) may be used off-label for SLE [36]. If participants go on biologic therapies during the study this will be explored as a confounding factor in the statistical analysis.

All participants must meet at least any ONE of the following criteria: (a) history of rheumatologist-documented clinical disease flare in the last year; (b) documentation of an SLE related emergency room visit in the last year; (c) history of rheumatologist-documented measure of active disease in the last year (e.g. organ system involvement due to SLE or Systemic Lupus Erythematosus Disease Activity Index (SLEDAI) score >3) AND history of steroid or other immune modifying drug used in last year (hydroxychloroquine, methotrexate, azathioprine, MMF, cyclophosphamide, etc.). In addition, participants must be able to understand the requirements of the study, provide written informed consent, including consent for the use and disclosure of research-related health information, and consent to comply with the study data collection procedures.

Exclusion criteria include unwillingness to wear the smartwatch, inability to understand the requirements of the study or provide written informed consent (including consent for the use and disclosure of research-related health information), inability to comply with the study visit schedule and procedures, or exclusion at the discretion of rheumatologist due to disease severity or ongoing pregnancy.

2.4. Study tasks and assessments

Participation in the study will be 12 months, subject to scheduling. Participants complete a clinical assessment at baseline, 3-, 6-, and 9-months at their respective enrollment sites. Participants will continue the digital portion of the study in months 9–12 but no 12-month clinical visit will occur. This study utilizes a comprehensive SLE focused platform employing both a novel blood-based biomarker index and a digital component consisting of a patient-centered app, a wearable smartwatch, and access to SLE-specific health coaching.

The rheumatologists will complete a start-of-study survey for each participant. This survey has 16 questions that address the participants' disease activity, SLE clinical management, and evaluation of the participants' self-management capability. The rheumatologist at each study site will complete an SLE PGA at each clinic visit as a reflection of overall disease severity and activity; the same provider will administer the SLE PGA at each visit [37]. The PGA was chosen due to its relatively low level of administrative burden [38] compared to the SLEDAI [39] and SLICC/ACR damage index [40]. Study participants will also be appropriately consented to obtain complete medical records for review and estimation of the SLEDAI [41] and SLICC/ACR damage index [42]. The participants will complete a start-of-study survey with 25 questions that collect general demographic information and address communication with their rheumatologist and care team, experience with SLE flares, and use of other resources.

The Lupus Flare Risk Index (L-FRI) applies a proprietary algorithm informed by immune biomarkers, outputting a single score estimating the risk of the patient experiencing a clinical disease flare in the next 12 weeks. The index was developed based on research showing that specific immune pathway alterations precede changes in clinical disease activity well before an impending SLE clinical flare [43,44]. Candidate soluble mediators including innate and adaptive cytokines, chemokines, as well as soluble receptors and TNF superfamily members were analyzed to develop a soluble mediator score (biomarker risk index) predictive of impending flare in both European American [43] and African American [44] SLE patients. Subsequent studies refined and narrowed the number of informative biomarkers to ultimately develop a refined biomarker risk index incorporating 11 blood plasma biomarkers [45]. The L-FRI was developed by comparison of log-transformed and standardized SLE-associated plasma mediator levels in samples from SLE patients with an imminent SELENA-SLEDAI Flare Index (SFI)-defined [46] clinical disease flare (Preflare) in the next 12 weeks vs. samples from demographically matched SLE patients without a flare over the same corresponding period (Pre-nonflare), weighted by their correlation to disease activity [47] at the time of concurrent Flare/Nonflare [45]. Following application of decision curve analysis to define low, medium and high scores, the refined L-FRI with 11 mediators exhibited a maximum sensitivity of 97 % and maximum specificity of 98 %. The plasma mediators included in the L-FRI are representative of several major immune pathways including: IL-4, IL-5 (Th-2-type); IL-17A (Th17-type); OPN (Th1- and Th17-type); MCP-1/CCL2, MCP-3/CCL7 (IFN-mediated chemokines); IL-7 (Homeostasis) and TNF superfamily members: BLyS (B-lymphocyte activation), TNF-α, TNFRI, and TNFRII (inflammatory cell activation). The L-FRI has been validated in an independent sample cohort [48].

Participants will complete four standardized and validated PRO questionnaires to assess SLE patient self-reported observation of his or her own condition and its impact on day-to-day life experience. Overall, these instruments allow participants to report their indexed ratings of the status and rate of change in status across multiple experiential and emotional categories. These PROs will be collected digitally and transmitted using a HIPAA compliant e-PRO such as JotForm.

The Self Efficacy for Managing Chronic Disease is a six-item instrument that measures how confident an individual is in engaging in health management activities including controlling fatigue, physical discomfort, emotional distress, and communicating with doctors [49]. Responses are provided on a 10-point scale ranging from 1 (not at all confident) to 10 (totally confident). A score is calculated for the instrument with a higher score indicating greater self-efficacy. The instrument was validated on 605 individuals with diverse chronic conditions and demonstrated an internal consistency reliability of .91 [49]. This instrument takes less than 5 min to complete.

The SLE Quality of Life Questionnaire (SLEQOL) is an SLE-specific 40-item instrument with a seven-point Likert scale used to evaluate the quality of life specific to SLE patients [50,51]. It has six major domains addressing physical functioning, activities, symptoms, treatment, mood, and self-image. A total score is calculated by summing all the responses giving a possible range of 40–280 and higher scores indicate poor quality of life. This instrument has been validated in several studies and is receiving broad utilization to quantify quality of life changes specifically in SLE patients [50,51]. SLE-QoL takes approximately 15–20 min to complete.

The Functional Assessment of Chronic Illness Therapy-Fatigue (FACIT-F) is a self-administered, widely utilized fatigue questionnaire that has been validated and utilized in SLE patients [52,53]. It has 13 questions with a five-point Likert scale that addresses several areas of fatigue including physical, functional, emotional and social consequences. The responses are added to give a total score that ranges from 0 to 52 with lower scores being associated with more fatigue. The approximate time for patients to complete this measure is estimated to be five to 7 min.

The Lupus Impact Tracker™ is a brief patient-completed questionnaire that has been shown to be reliable and valid for assessing and monitoring the impact of SLE [54]. The LIT has 10 questions addressing cognition, SLE medications, physical health, pain/fatigue impact, emotional health, body image, and planning/desires/goals. The responses are scaled from 0 (none of the time) to 4 (all of the time) and are added to arrive at a total raw score ranging from 0 to 40. This raw score is located on a chart with scores weighted on a 0 to 100 score to determine the Lupus Impact score. Lower scores are associated with less impact of SLE on life. The LIT takes at most three to 5 min to complete.

Following their 3-month clinic visit, participants will be given a Withings Steel-HR smartwatch and a unique code to access the study mobile phone app [55]. This smartwatch has clinically validated capabilities and has been used in several published studies [56–58]. It is a durable device that is convenient to wear continuously due to the

battery life lasting up to 30 days before requiring recharging. Together, the mobile phone app and the smartwatch simplify continuous measurement and collection of physiological data.

Participants will be guided through onboarding by (a) creating an account, (b) consenting to the terms of the app, and (c) scheduling an introductory session with an SLE-specific health coach. Upon completion of onboarding, the participant will be guided through educational modules on how to use the app, track symptoms, and communicate with their health coach. They will be instructed to wear the smartwatch both day and night to automatically measure and record their daily physical activity (steps and distance), heart rate, and sleep patterns through integration into the study mobile phone app. The participants will be asked to wear the smartwatch through the end of month 12 of the study.

Health coaching specific to SLE will be offered as part of the program. When participants are enrolled in the mobile phone app component, they will be encouraged to schedule an initial virtual session with a health coach to familiarize them with the independent courses available in the health coaching modules and the process for scheduling video sessions with the health coach.

The focus of health coaching is on SLE and is designed to provide a bridge to inform, support, and educate participants between clinic visits with the goal of easing the burden of managing their symptoms. Health coaches will be required to be a National Board-Certified Health & Wellness Coach with at least one year's experience coaching patients with SLE and/or other autoimmune conditions. Adherence to the program will be monitored through health coach reports on the completed session with the goal of each participant completing two sessions per month.

The purpose of the health coaching sessions will be to provide an environment for the participant to learn, grow, set, and accomplish goals in the areas of fitness and supportive exercises, health literacy, nutrition, wellness, and management of several SLE related challenges (e.g. energy, fatigue, sleep, pain, stress, flare, brain fog). Specifically, the following modules are available in the mobile app: (a) understanding SLE; (b) goal setting; (c) energy and fatigue management; (d) sleep management; (e) pain management; (f) stress management; (g) flare management; (h) brain fog management; (i) wellness support; (j) nutrition support; (k) fitness support; (l) emotional, mental, and spiritual support; (m) personal and professional relationship support; (n) general empowerment and support; (o) fitness and exercise; (p) yoga.

The final participants' end of study survey will be conducted in month 12. This survey has 16 questions and addresses the participants' experience in the study with the L-FRI blood test, the app, the wearable watch, and the health coaching. The participants will also have the opportunity to complete a semi-structured one-on-one interview regarding the study. These interviews will be conducted by an independent consultant using a HIPAA compliant video meeting platform. This interview will consist of 22 questions addressing each component of the comprehensive program; the purpose is to collect more in-depth information on the participants' study experience.

Rheumatologists will complete an end-of-study survey with 18 questions addressing SLE, patient self-management, and their overall experience and value of the comprehensive program for managing the study participants at their site. An end-of-study focus group session with all five site rheumatologists will be conducted by an independent consultant using a discussion guide with 30 questions that addresses their use and value of the comprehensive program, the L-FRI blood test, biometric data collected by the wearable, PRO symptom data, and health coaching.

2.5. Major analyses and methods

2.5.1. Analysis of the primary objective

The primary objective of the study will be to analyze the within-person change in total scores on the Self-Efficacy for Managing Chronic Disease Questionnaire between pre-intervention (Baseline and Month 3) compared to post-intervention (months 4–12) periods. The mean of the within-person total scores in the pre-intervention period will be compared with the corresponding mean for the intervention period. Within-individual (same participant) longitudinal differences across the full study from baseline to 12 months will also be analyzed. Statistical methods are discussed below. We will evaluate the effect of the comprehensive platform on patient health self-efficacy, defined as an individual's confidence in managing their fatigue, physical discomfort, emotional distress, and communication with doctors.

The responses provided on the additional PRO surveys will be analyzed to provide additional detail and insight into each participant's SLE condition. In particular, analysis of the SLEQOL score for each of the six major domains will be used to determine specific areas of potential impact including physical functioning, activities, symptoms, treatment, mood, and self-image [50,51]. The LIT is also SLE-specific and queries medications, physical health, pain/fatigue impact, emotional health, body image, and planning/desires/goals [54]. The FACIT-F specifically addresses several areas of fatigue including physical, functional, emotional and social consequences [52,53]. Analyses of this additional survey data will be performed as discussed above.

2.5.2. Analysis of the secondary objective

The secondary objective is to evaluate the usefulness and impact of the comprehensive platform for rheumatologists in managing patients with SLE. The rheumatologists will have completed a start-of-study survey for participants with 15 questions on which they rate the participants' disease activity, SLE clinical management, and evaluation of the participants' self-management capability. At the end of the study, they will complete questionnaires with additional questions asking them to rate the extent to which the program is helpful in managing SLE as well as whether communication and interactions with their patients improved during the study. Analyses will compare responses from the start of the study to the end of the study to determine how useful they found the program for managing SLE cases.

2.5.3. Exploratory analyses

The exploratory objective is to determine if rheumatologists' perceptions of their patients' clinical status as measured by the PGA, medical record estimation of the SLEDAI [41] and SLICC/ACR damage index [42], and rheumatologists' surveys correlate with patient's measured immunological status as determined by their L-FRI index scores and/or individual biomarker values. Participants' study and medical records will be analyzed to determine if changes in SLE symptoms measured by PGA as well as estimated SLEDAI and SLICC/ACR damage index correlate with biomarker results. Exploratory analyses will also be performed to examine the potential relationship between biometric parameters, the L-FRI score, and individual biomarker levels. One published study examining the relationship between heart rate variability (HRV), disease activity, and alteration of certain soluble mediator pathways has demonstrated that impaired HRV was associated with SLE disease activity and levels of select plasma IFN type II and TNF pathway mediators [39].

2.5.4. Statistical methods

For continuous variables, descriptive statistics such as the number of observations, mean, and standard deviation will be presented [59]. Data will be tested for normal distribution. For categorical variables, descriptive statistics, including frequencies and percentages, will be tabulated. Statistical tests, specifically Student's t-test for continuous measures and Chi-square for categorical measures, will be employed to assess the difference between severity groups, with a two-tailed significance level of .05 utilized and related p-values reported. Multiple comparison p-values will be corrected using Bonferroni correction.

Partitioning the full metric basis into outcome terms (***Y***: the superset of SLE-symptom-instances, co-pathological occurrences, and PRO negative impacts) versus situational factors (***X***: therapeutic regimens,

healthcare utilization, Withings records, PRO self-efficacy terms, and health coach visits) will enable assessment of interdependence relationships. For static relationships, Feller auto-covariance, Pearson product-moment correlation coefficients and Weisstein cross-correlation matrices will be assembled, both on a patient-whole-study basis and at a patient-quarterly level. If data are not normally distributed, then Spearman non-parametric analyses will be used instead of Pearson. For outcome metrics $\boldsymbol{y} \varepsilon \boldsymbol{Y}$ with prospectively significant relationships ($p < 0.10$) to situational metrics $\boldsymbol{x} \varepsilon \boldsymbol{X}$, conditionality distributions will be computed, and Granger causality tests will be attempted, using lags of length up to 1 quarter and potentially ranging down (if data is adequate) to lengths of 1–2 months. All calculations will be performed using the Statsmodels Module for Python (Seabold, Skipper, and Josef Perktold. "Statsmodels: Econometric and statistical modeling with Python." Proceedings of the 9th Python in Science Conference. 2010. Updated for Python 3.12).

3. Results

Enrollment started in June of 2022 and continued over the next 16 months. All assessments were completed in October of 2024 with final medical record collection completed in January 2025. Study data analysis and reports are expected in March of 2025.

4. Discussion

SLE patients and their rheumatologists face a myriad of challenges in disease management. The heterogeneity and complexity of symptoms faced by an individual with SLE can be daunting. One effective approach to overcome these barriers is self-management programs focused on patient education, improved communication with their providers, and interaction with a supportive certified health coach. It has been demonstrated that individuals with chronic diseases who are actively involved in these types of programs experience improved health outcomes [60–63]. However, patient participation has been reported to be poor when *real time, in-person* group attendance is required. Innovative approaches, such as those being utilized in the current study, have demonstrated that web- and mobile phone-based applications are beneficial for disease management through patient education (especially in rural communities) and promote better treatment compliance [32,33]. In SLE disease management this comprehensive self-management program has the potential to be beneficial in filling gaps between in-person clinic visits when patients are often lost to follow-up due to the current and predicted ongoing shortage of rheumatologists [14]. This program would allow for monitoring of the patient's condition to effectively manage those with stable disease and prioritize clinic appointments for those at highest risk of heightened disease activity and flare.

There is currently a knowledge gap in the application of digital technologies to provide real time, objective behavioral and physiological data within the context of SLE disease management and specifically disease activity and flare. The biomarker risk index and digital data from connected wearable devices and smartphone apps offer a novel avenue to explore the complex and dynamic nature of SLE disease activity and flare in real time. The results of this pilot study will provide valuable insights into the application of this comprehensive SLE management platform to standard of care approaches and contribute to the design of a larger trial to confirm the impact on disease management and outcomes.

CRediT authorship contribution statement

Eldon R. Jupe: Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Mohan Purushothaman:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Funding acquisition, Conceptualization. **Bruce Wang:** Writing – review & editing, Writing – original draft, Methodology. **Gerald Lushington:** Writing – review & editing, Writing – original draft, Methodology. **Sandeep Nair:** Writing – review & editing, Software, Project administration, Methodology. **Vijay R. Nadipelli:** Writing – review & editing, Visualization, Project administration, Methodology, Conceptualization. **Bernard Rubin:** Writing – review & editing, Visualization, Project administration, Methodology, Conceptualization. **Melissa E. Munroe:** Writing – review & editing, Methodology, Conceptualization. **Jessica Crawley:** Writing – review & editing, Project administration, Methodology. **Sneha Nair:** Writing – review & editing, Project administration. **Georg Armstrong:** Writing – review & editing, Project administration, Methodology. **Arif Sorathia:** Writing – review & editing, Visualization, Project administration, Methodology, Funding acquisition, Conceptualization. **Brett Adelman:** Writing – review & editing, Visualization, Project administration, Methodology, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Eldon Jupe, Mohan Purushothaman, Sandeep Nair, Melissa Munroe, Jessica Crawley, Sneha Nair, Georg Armstrong, Arif Sorathia, and Brett Adelman are or were employed by Progentec Diagnostics, Inc. during the time this work was performed. Bruce Wang and Gerald Lushington are paid consultants to Progentec. Vijay Nadipelli is a full-time employee with GSK with stock ownership. Bernard Rubin is currently employed by Progentec and was previously a full-time employee with GSK with stock ownership.

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Data availability

No data was used for the research described in the article.

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